

★ LEVEL 2

 IQCDL CERTIFICATION PROGRAM

Practitioner Level

Quantum Computing & Post-Quantum Cryptography

مستوى الممارس - الحوسبة الكمية وتشفير ما بعد الكم



5 Days

Intensive Training



Hands-On

Practical Coding



150 MCQ

Certification Exam



Certified

Practitioner

Program Overview

نظرة عامة على البرنامج

Program Description

وصف البرنامج

The Practitioner Level is an intensive 5-day hands-on program designed for technical professionals who want to implement quantum computing solutions and post-quantum cryptography in real-world scenarios.

Prerequisites

المتطلبات الأساسية

Required: Completion of Foundation Level certification and basic programming knowledge (Python recommended).

Recommended: Familiarity with cryptography concepts and software development experience.

Certification Exam

امتحان الشهادة

150

Questions | أسئلة

240

Minutes | دقيقة

5-Day Schedule

جدول الخمسة أيام

1

Quantum Programming Fundamentals

Setup quantum development environment, learn Qiskit basics, implement quantum circuits.

2

Advanced Quantum Algorithms

Implement Shor's factoring and Grover's search algorithms, quantum error correction.

3

PQC Implementation

Hands-on implementation of CRYSTALS-Kyber, Dilithium, and SPHINCS+ with benchmarking.

4

Hybrid Cryptographic Solutions

Design hybrid classical-PQC systems, implement TLS with PQC, crypto-agile architectures.

5

Migration Planning & Case Studies

Quantum Programming Fundamentals

أساسيات البرمجة الكمية



Learning Objectives

أهداف التعلم



Setup Development Environment: Install and configure Qiskit, Python, Jupyter Notebook, and quantum simulators



Learn Qiskit Basics: Understand quantum registers, classical registers, quantum circuits, and basic operations



Build Quantum Circuits: Create and manipulate quantum circuits using gates, measurements, and visualizations



Visualize Quantum States: Use Bloch sphere, statevector, and histogram visualizations to understand quantum behavior



Run Simulations: Execute quantum circuits on local simulators and analyze measurement results

Development Tools

أدوات التطوير

Quantum Gates and Circuits

البوابات الكمية والدوائر

Essential Quantum Gates

البوابات الكمية الأساسية

X Gate

Pauli-X (NOT): Flips $|0\rangle \leftrightarrow |1\rangle$. Equivalent to classical NOT gate. Creates bit-flip.

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

H Gate

Hadamard: Creates superposition. Transforms $|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$. Most fundamental gate.

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Z Gate

Pauli-Z: Phase flip. Leaves $|0\rangle$ unchanged, flips phase of $|1\rangle$ to $-|1\rangle$.

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Circuit Construction

بناء الدوائر

</> Bell State Circuit

```
# Create entangled pair qc = QuantumCircuit(2, 2)
qc.h(0) # Hadamard on q0 qc.cx(0, 1) # CNOT: q0-q1
qc.measure_all()
```

</> Superposition Example

```
# Equal superposition qc = QuantumCircuit(3)
qc.h([0, 1, 2]) # H on all # Result: 8 states
equally
```

💡 Key Concepts | المفاهيم الأساسية



Qiskit Hands-On Practice

Qiskit التدريب العملي على

Example 1: Bell State Circuit

مثال 1: دائرة حالة بيل

```
# Import Qiskit libraries from qiskit import QuantumCircuit,
transpile from qiskit_aer import AerSimulator # Create a quantum
circuit with 2 qubits qc = QuantumCircuit(2, 2) # Apply Hadamard
gate to qubit 0 qc.h(0) # Apply CNOT gate (control=0, target=1)
qc.cx(0, 1) # Measure both qubits qc.measure([0, 1], [0, 1]) #
Simulate the circuit simulator = AerSimulator() compiled_circuit =
transpile(qc, simulator) result = simulator.run(compiled_circuit,
shots=1024).result() counts = result.get_counts() print(counts) #
{'00': ~512, '11': ~512}
```

Understanding the Code

فهم الكود

- 1 Import Libraries:** Import Qiskit core modules for circuit creation and simulation.
- 2 Create Circuit:** Initialize a quantum circuit with 2 qubits and 2 classical bits.
- 3 Apply Gates:** Hadamard gate creates superposition, CNOT creates entanglement.
- 4 Measure:** Collapse quantum states to classical bits for observation.
- 5 Simulate:** Run the circuit 1024 times and collect measurement results.

Expected Output | المخرجات المتوقعة:

{'00': 512, '11': 512} This creates a maximally entangled Bell state where measuring one qubit instantly determines the other's state.

Example 2: Quantum Superposition

مثال 2: التراكب الكمي

```
# Create a circuit with 1 qubit qc = QuantumCircuit(1, 1) # Apply
Hadamard gate to create superposition qc.h(0) # Measure the qubit
qc.measure(0, 0) # Visualize the circuit
print(qc.draw(output='text')) # Simulate and get results simulator
```

Key Concepts

المفاهيم الأساسية

- 1 Superposition:** The qubit exists in both $|0\rangle$ and $|1\rangle$ states simultaneously until measured.
- 2 Equal Probability:** After Hadamard gate, 50% chance of measuring 0 or 1.

Advanced Quantum Algorithms

الخوارزميات الكمية المتقدمة



Shor's Algorithm

خوارزمية شور للتحليل العائلي



Purpose | الغرض

Factor large integers exponentially faster than classical algorithms, breaking RSA encryption



Complexity | التعقيد

$O((\log N)^3)$ quantum vs $O(e^{(1.9(\log N)^{1/3})})$ classical



Implementation | التنفيذ

Quantum Fourier Transform, period finding, modular exponentiation



Grover's Algorithm

خوارزمية جروفر للبحث



Purpose | الغرض

Search unsorted databases with quadratic speedup over classical search



Complexity | التعقيد

$O(\sqrt{N})$ quantum vs $O(N)$ classical for N items



Implementation | التنفيذ

Oracle construction, amplitude amplification, diffusion operator



Impact | التأثير

Reduces effective key length: AES-256 → AES-128 security

Shor's Algorithm Implementation

Integer Factorization - تنفيذ خوارزمية شور

Grover's Algorithm Implementation

Quantum Search - تنفيذ خوارزمية جروفر

Key Concepts

المفاهيم الأساسية



Oracle

Black box function that marks the target state by flipping its phase: $|x\rangle \rightarrow -|x\rangle$ if x is the solution



Amplitude Amplification

Increases probability amplitude of marked state while decreasing others through repeated iterations



Diffusion Operator

Inverts amplitudes about the average: $2|\psi\rangle\langle\psi| - I$, where $|\psi\rangle$ is uniform superposition



Iterations

Repeat Oracle + Diffusion approximately $\pi/4 \times \sqrt{N}$ times for optimal probability

Qiskit Implementation

Qiskit التنفيذ باستخدام

```
from qiskit import QuantumCircuit
from qiskit.algorithms import Grover
from qiskit.circuit.library import PhaseOracle

# Define oracle for target state '11'
oracle = PhaseOracle('x0 & x1')

# Create Grover's algorithm instance
grover = Grover(oracle=oracle)

# Build the circuit
circuit = grover.construct_circuit(
    iterations=1, #  $\pi/4 \times \sqrt{4} \approx 1$ 
    measurement=True
)

# Execute and get results
result = backend.run(circuit, shots=1024).result()
counts = result.get_counts()
print(counts) # {'11': ~1024}
```


Post-Quantum Cryptography Implementation

تنفيذ تشفير ما بعد الكم

Learning Objectives

أهداف التعلم

-  **Understand NIST Standards:** Learn the three NIST-approved PQC algorithms and their standardization process (FIPS 203, 204, 205)
-  **Implement CRYSTALS-Kyber:** Build key encapsulation mechanism (KEM) for secure key exchange using lattice-based cryptography
-  **Implement CRYSTALS-Dilithium:** Create digital signature schemes resistant to quantum attacks using module lattices
-  **Implement SPHINCS+:** Build stateless hash-based signature schemes with minimal security assumptions
-  **Performance Benchmarking:** Compare PQC algorithms with classical cryptography in terms of key size, speed, and security

NIST-Approved PQC Algorithms

NIST المعتمدة من PQC خوارزميات



CRYSTALS-Kyber (ML-KEM)

FIPS 203 | Key Encapsulation Mechanism

Lattice-based KEM for secure key exchange. Fast, small keys, and high security. Recommended for TLS, VPN, and secure communications.



CRYSTALS-Dilithium (ML-DSA)

FIPS 204 | Digital Signature Algorithm

Lattice-based digital signatures. Efficient signing and verification. Ideal for certificates, code signing, and authentication.



SPHINCS+ (SLH-DSA)

FIPS 205 | Stateless Hash-Based Signatures

CRYSTALS-Kyber (ML-KEM)

Module-Lattice-Based Key Encapsulation Mechanism | آلية تغليف المفاتيح القائمة على الشبكات



Algorithm Overview

نظرة عامة على الخوارزمية

Security Level Kyber-512	Equivalent AES-128
Public Key 800 bytes	Ciphertext 768 bytes
Security Level Kyber-768	Equivalent AES-192
Security Level Kyber-1024	Equivalent AES-256

Key Features

Python Implementation

Python التنفيذ باستخدام

```
# Install: pip install pqcrypto from pqcrypto.kem.kyber512 import
generate_keypair, encrypt, decrypt # Step 1: Generate key pair
(Alice) public_key, secret_key = generate_keypair() # Step 2:
Encapsulate shared secret (Bob) ciphertext, shared_secret_bob =
encrypt(public_key) # Step 3: Decapsulate shared secret (Alice)
shared_secret_alice = decrypt(secret_key, ciphertext) # Verify: Both
parties have same shared secret assert shared_secret_alice ==
shared_secret_bob print(f"Shared secret established:
{len(shared_secret_alice)} bytes") # Output: Shared secret
established: 32 bytes
```

Use Cases

حالات الاستخدام



CRYSTALS-Dilithium (ML-DSA)

Module-Lattice-Based Digital Signature Algorithm | خوارزمية التوقيع الرقمي القائمة على الشبكات

🌟 NIST FIPS 204 Standard

Security Parameters

معاملات الأمان

SECURITY LEVEL

2, 3, 5

NIST Categories

PUBLIC KEY

1312B

Dilithium3

SIGNATURE

2420B

Dilithium3

SPEED

Fast

Sign & Verify

Python Implementation Example

Python + liboqs

```
from oqs import Signature # Initialize Dilithium3
signer = Signature("Dilithium3") # Generate key pair
public_key = signer.generate_keypair()
secret_key = signer.export_secret_key() # Sign a message
message = b"Important document"
signature = signer.sign(message)
print(f"Public key size: {len(public_key)} bytes")
print(f"Signature size: {len(signature)} bytes") # Verify signature
verifier = Signature("Dilithium3")
is_valid = verifier.verify(message, signature, public_key)
if is_valid: print("✓ Signature verified successfully")
else: print("✗ Signature verification failed")
```

Key Features

الميزات الرئيسية

Use Cases | حالات الاستخدام

Hybrid Cryptographic Solutions

الحلول التشفيرية الهجينة



Learning Objectives

أهداف التعلم



Design Hybrid Systems: Combine classical and PQC algorithms for backward compatibility and enhanced security



Implement TLS with PQC: Integrate post-quantum algorithms into TLS handshakes and secure communications



Build Crypto-Agile Architecture: Design systems that can quickly switch between cryptographic algorithms

Hybrid Approaches

النهج الهجينة



Concatenation

Use both classical and PQC algorithms together for dual protection



Nested

Encrypt with one algorithm, then encrypt the result with another



Parallel



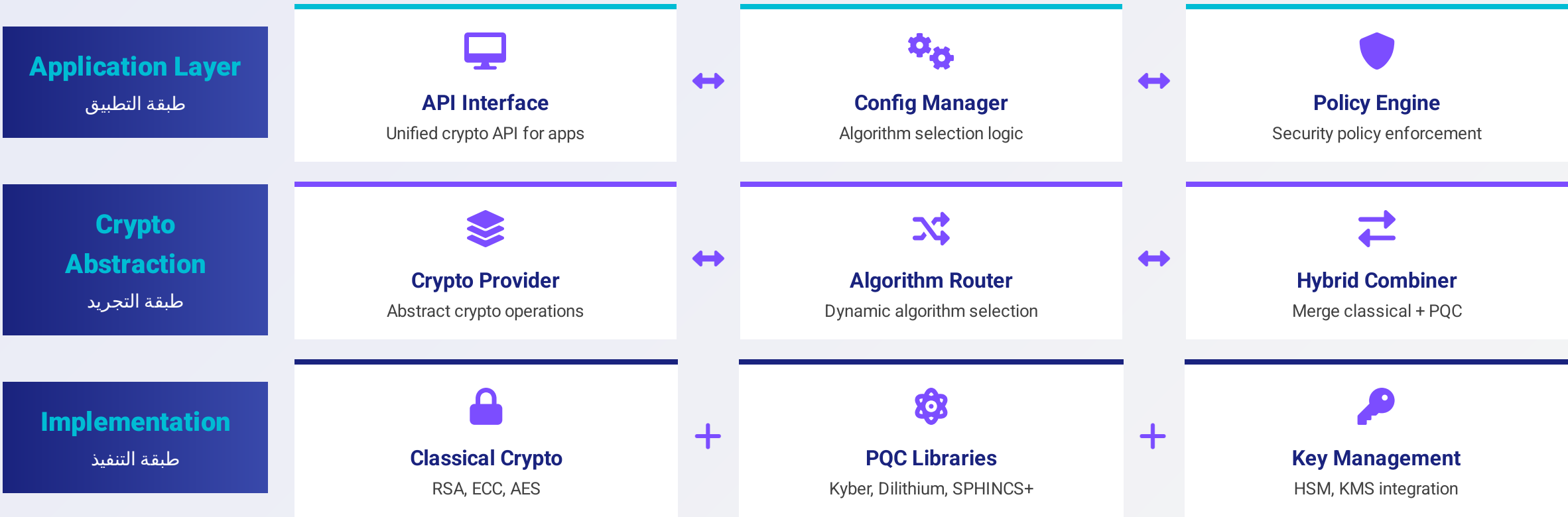
Fallback

Hybrid Crypto-Agile Architecture

البنية المرنة للتشفير الهجين

Layered Architecture Design

تصميم البنية متعددة الطبقات



Design Principles for Crypto-Agility

TLS with Post-Quantum Cryptography Integration

TLS دمج تشفير ما بعد الكم في بروتوكول

Hybrid TLS Handshake Flow

الهيئة TLS تدفق مصافحة

1

Client Hello

Client sends supported cipher suites including both classical (ECDHE) and PQC (Kyber) key exchange algorithms



2

Server Hello

Server selects hybrid mode: X25519 + Kyber-768 for key exchange, Dilithium for certificate signatures



3

Certificate Exchange

Server presents X.509 certificate signed with Dilithium. Client verifies using PQC signature verification



4

Key Exchange

Both parties perform X25519 and Kyber key exchange. Shared secrets are concatenated and hashed together



5

Session Established

Master secret derived from combined keys. AES-256-GCM used for symmetric encryption. Connection secured against quantum attacks

OpenSSL Configuration Example

OpenSSL مثال تكوين

```
# Install OQS-OpenSSL fork git clone https://github.com/open-quantum-safe/openssl.git cd openssl && ./config && make && make install # Generate PQC certificate openssl req -x509 -new -newkey dilithium3 \ -keyout server_dilithium.key \ -out server_dilithium.crt -nodes \ -subj "/CN=example.com" -days 365 # Configure TLS server with hybrid mode openssl s_server -cert server_dilithium.crt \ -key server_dilithium.key \ -curves "kyber768:X25519" \ -www -port 443 # Test connection with hybrid client openssl s_client -connect localhost:443 \ -curves "kyber768:X25519"
```

Security Benefits | فوائد الأمان



Quantum-Safe
Key Exchange



PQC Digital
Signatures



Backward
Compatible

PROTOCOL

TLS 1.3

KEY EXCHANGE

X25519 + Kyber-768

SIGNATURE

Dilithium3

SYMMETRIC

AES-256-GCM

SECURITY LEVEL

NIST Level 3

PQC Migration Planning

تخطيط الانتقال إلى تشفير ما بعد الكم



Migration Roadmap

خارطة طريق الانتقال

1

PHASE 1

Discovery & Inventory | الاكتشاف والجرد

Identify all cryptographic assets, algorithms, and dependencies. Document current encryption methods, key management systems, and certificate infrastructure.

2

PHASE 2

Risk Assessment | تقييم المخاطر

Evaluate quantum threat timeline using Mosca's Theorem. Prioritize critical systems based on data sensitivity, shelf life, and compliance requirements.

Pilot Implementation | التنفيذ التجريبي

Implementation Checklist

قائمة التنفيذ



Cryptographic Inventory: Map all encryption points, certificates, keys, and algorithms in use



Stakeholder Engagement: Involve IT, security, compliance, and business teams in planning



Vendor Coordination: Ensure suppliers support PQC algorithms and migration timelines



Testing Framework: Establish comprehensive testing for functionality, performance, and compatibility



Documentation: Maintain detailed records of changes, configurations, and rollback procedures

CBOM - Cryptographic Bill of Materials

قائمة جرد المواد التشفيرية

What is CBOM?

ما هو CBOM؟

A comprehensive inventory of all cryptographic assets, algorithms, keys, and certificates used across an organization's infrastructure. Essential for identifying quantum-vulnerable components and planning migration to PQC.

Sample Cryptographic Inventory

نموذج جرد تشفيري

System/Application	Algorithm	Key Size	Quantum Status
Web Server TLS	RSA	2048-bit	Vulnerable
VPN Gateway	ECDHE (P-256)	256-bit	Vulnerable
Database Encryption	AES-256	256-bit	Safe (with caution)
Code Signing	RSA-4096	4096-bit	Vulnerable
Email S/MIME	RSA-2048	2048-bit	Vulnerable
SSH Access	Ed25519	256-bit	Vulnerable
API Authentication	HMAC-SHA256	256-bit	Safe

CBOM Creation Process

عملية إنشاء CBOM

1

Discovery

Scan all systems, applications, and infrastructure to identify cryptographic implementations

2

Documentation

Record algorithm types, key sizes, usage contexts, and dependencies

3

Risk Assessment

Evaluate quantum vulnerability and prioritize based on data sensitivity

4

Classification

Categorize by migration priority: critical, high, medium, low

5

Maintenance

Continuously update inventory as systems change and new crypto is deployed

أدوات الاكتشاف | Discovery Tools

Real-World Case Studies

دراسات حالة من الواقع العملي



Banking

القطاع المصرفي

JPMorgan Chase - Quantum-Safe Network Deployment

CHALLENGE

Protect financial transactions and customer data from future quantum attacks. Legacy systems using RSA-2048 for transaction signing and TLS for secure communications.

SOLUTION

Deployed **hybrid TLS with Kyber-768 + ECDH** for key exchange and **Dilithium-3** for transaction signatures. Implemented crypto-agile architecture for future algorithm updates.

RESULTS

Successfully migrated interbank messaging system. Maintained backward compatibility with existing infrastructure. Achieved quantum-readiness by 2024 with minimal performance impact.



Telecom

الاتصالات

Deutsche Telekom - 5G Network Security with PQC

CHALLENGE

Secure 5G core network infrastructure against quantum threats. Protect subscriber authentication, network slicing, and edge computing communications with long-term security requirements.

SOLUTION

Integrated **CRYSTALS-Kyber** into 5G authentication framework. Deployed quantum-safe VPNs using hybrid IPsec. Collaborated with equipment vendors for firmware updates.

RESULTS

Pilot deployment in Munich covering 50,000 subscribers. Network latency increased by less than 2ms. Full rollout planned across European operations by 2026.



Healthcare

الرعاية الصحية

NHS Digital - Patient Data Protection Initiative

CHALLENGE

Protect sensitive patient records stored for decades. Medical data must remain confidential

SOLUTION

Implemented **hybrid encryption** with AES-256 + Kyber-1024 for data at rest. Deployed **SPHINCS+**

RESULTS

Protected 60 million patient records across England. Compliant with GDPR and UK data



You've Completed the IQCDL Practitioner Level Training

01

المستوى التأسيسي

🕒 3 Days Training

02

مستوى الممارس

🕒 5 Days Training

03

مستوى الخبير

🕒 Coming Soon